

What do we learn:-

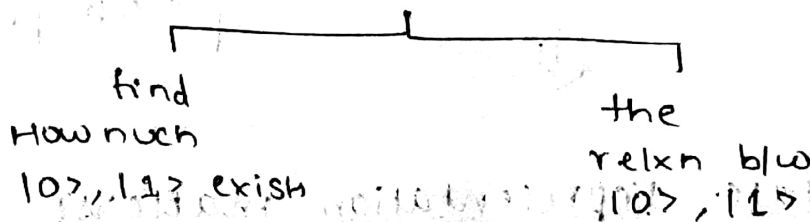
- Bloch sphere and state representation
- B-Sphere states
- M-Metaphors
- Real world analogy

State Representation:-

ex: • Qbit in Superpos

But where is the Qbit

Reason: Qbit has ∞ possible combinations



Q-state representation

m: Needle Compass

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

↓
amplitude of $|0\rangle$

amp. $|1\rangle$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Rule:

$$|\alpha|^2 + |\beta|^2 = 1$$

- if not followed, can't get third outcome

Ex:-

$$① |\psi\rangle = |0\rangle + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$P(|0\rangle) = 100\%$$

$$P(|1\rangle) = 0\%$$

$$|0\rangle \begin{array}{|c|c|c|c|} \hline \hline \hline \hline \hline \end{array}$$

$$|1\rangle \begin{array}{|c|c|c|c|} \hline \hline \hline \hline \hline \end{array}$$

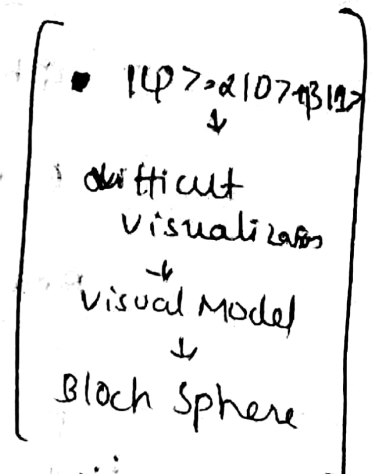
②

$$|\psi\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

↓

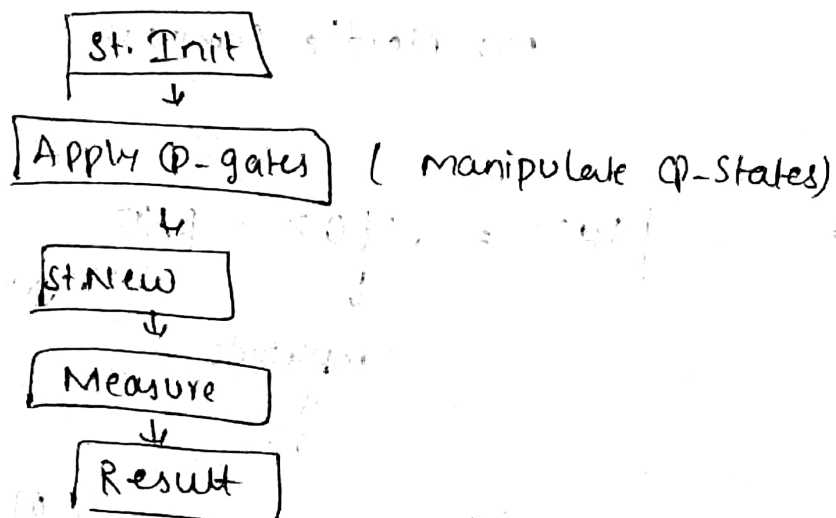
$$P(|0\rangle) = 50\% \rightarrow 50-50$$

(Equal Superposition)



Why does state representation matters?

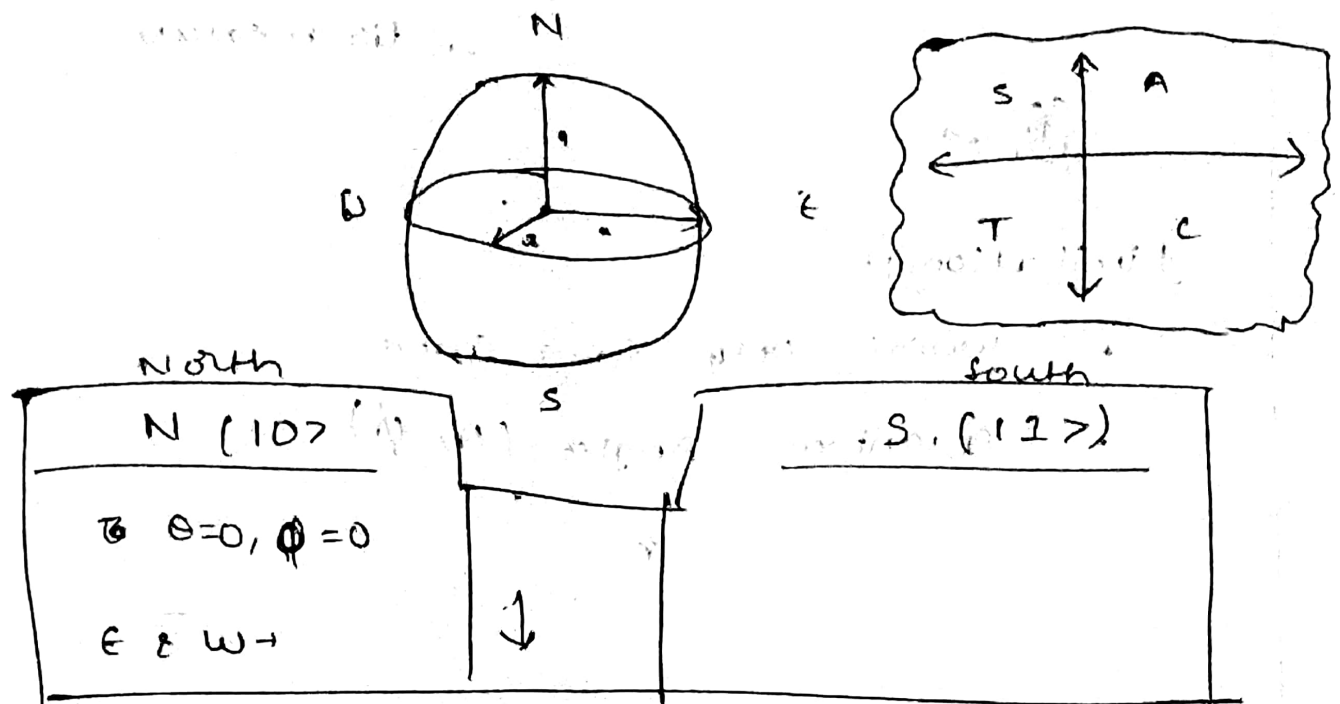
- Q-Algs don't directly manipulate bits
- Manipulate : Q-States



BLOCH SPHERE:-

→ 3D visualization of a single qubit

→ Every qubit corresponds from χ to surface of sphere



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2}$$

Z-Basis		X-Basis		Y-Basis	
N	S	R-side	L-side	Front	Back
$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	$ +\rangle$	$ -\rangle$
→ computational basic states → Final Measure		R		→ Introduce phase info	
		L			
		$ +\rangle = (0\rangle + 1\rangle) / \sqrt{2}$ $ -\rangle = (0\rangle - 1\rangle) / \sqrt{2}$			
		→ Heavily used in q-algs			

ex:-

Q-bit start at center

→
C-gate rotate arrow



Limitations:-

- works only for 1-qubit

- Quantum Angles (θ, ϕ)

Pros:- see,

→ state changes

→ superposition

→ Rotations

→ dis-measure

